

Unstructured/ Structured/ Semi-structured Data

Data isn't the same as information. Data is the raw facts. That means you should think of data in terms of the individual fields or columns of data you may find in a relational database or perhaps the loose document (tagged with some descriptors called metadata) stored in a document repository. On their own, these items are unlikely to make much sense to you or to a business. And that's perfectly okay — sometimes. Information is the collective body of all those data parts, that results in the factoids making logical sense.

Unstructured data is ambiguous, having no rhyme, reason, or consistency whatsoever. Pretend that you're looking at a batch of photos or videos. Are there explicit data points that one can associate with a video or photo? Perhaps, because the file itself may consist of a structure and be made of some metadata. However, the byproduct itself — the represented depiction — is unique. The data isn't replicable; therefore, it's unstructured. That's why any video, audio, photo, or text file is considered unstructured data

Structured data is, as it appears in a Microsoft **Excel spreadsheet**. Whether you're using Power BI for personal analysis, educational purposes, or business support, the most accessible data sources for BI tools are structured. Platforms that offer robust structured data options would include Microsoft SQL Server, Microsoft Azure SQL Server, Microsoft Access, Azure Table Storage, Oracle, IBM DB2, MySQL, PostgreSQL, Microsoft Excel, and Google Sheets.

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Semi structured data does have some formality, but it isn't stored in a relational system and it has no set format. Instead, semi structured data contains tags that make the data easier to organize in some form of hierarchy. Nonrelational data systems or NoSQL databases are best associated with semi structured data, where the programmatic code, often serialized, is driven by the technical requirements. There is no hard-and-fast coding practice. For the business intelligence developer utilizing semi structured languages, serialized programming practices can assist in writing sophisticated code. Whether the goal is to write data to a file, send a data snippet to another system, or parse the data to be translatable for structured consumption, semi structured data does have the potential for business intelligence systems. If the serialized language can communicate and speak the same language, a semi structured dataset has great potential.
